Functional dependency

$\alpha \xrightarrow{\text{determines}} \beta$

2.1 → 5.2

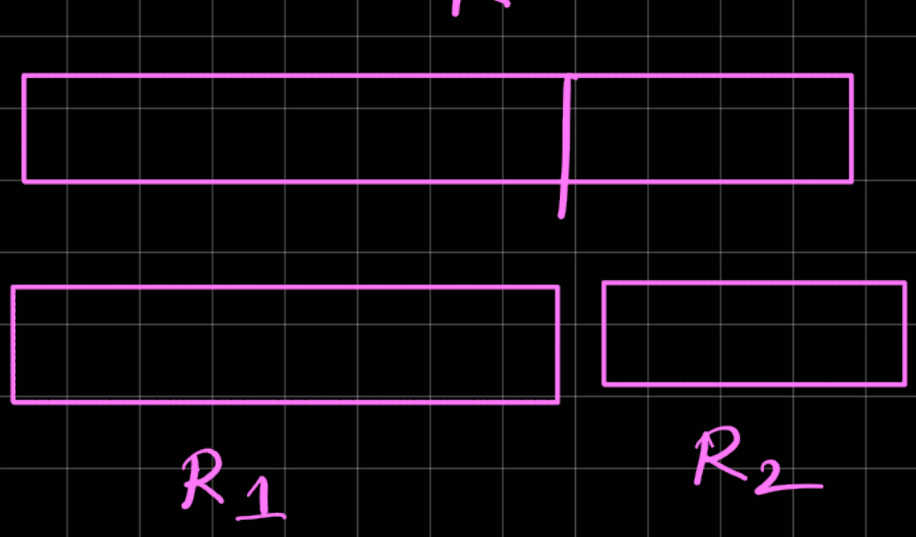
if we get 2.1 guaranted to get 5.2

not vice versa.

2.3 → 5.2 ✓

3.1 → 5.1

2.1 → 5.2 ✓



splitting → schema decomposition.

F

$$\begin{bmatrix} \alpha \rightarrow \beta \\ \beta \rightarrow \delta \\ \delta \rightarrow C \end{bmatrix}$$

$\alpha \rightarrow \delta$ (Rule of Transitivity)

F^+

$$\begin{bmatrix} \alpha \rightarrow \beta \\ \beta \rightarrow \delta \\ \delta \rightarrow C \\ \alpha \rightarrow \delta \end{bmatrix}$$

R | EID | Name | DOB | ϕ N | PID (issue) | PED (Expire)

TCP IP → 2 s, 4 s, 8 s double up waiting time.

Most efficient.

NID

CID	CName	DOB	PN

↓ FK

Passport

PN	PID	PEN

$R_1 \cap R_2 = PN$ PN → passport

NID

CID	CName	DOB

Passport

CID	PN	PID	PEN

$R_1 \cap R_2 = CID$ CID → (NID)

CID → CID, CName, DOB

101 → 101, a, 1-1-91

102 → 102, b, 1-1-92

lossless Decomp.